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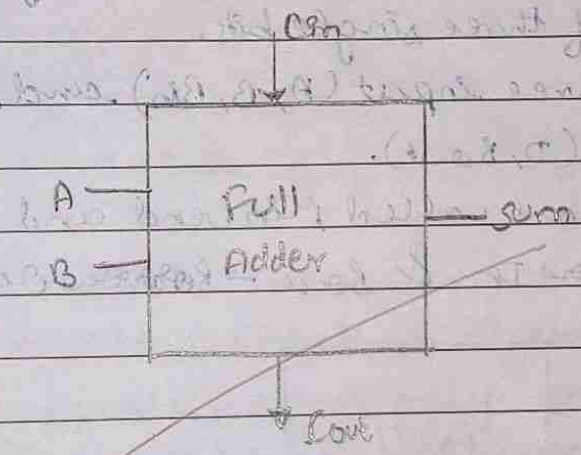
Assignment - ①

Unit 1:- Combinational Logic Design

(Q1) Design full adder and full substractor - {6 M}

⊙ Full adder:-

1) A full adder is a combinational circuit that forms the arithmetic sum of three input bits.
2) It consists of three inputs and two outputs.
3) Two of the input variables, denoted by A & B, represents the two significant bits to be added. The third input Cin, represent the carry from the previous lower significant position.



Block schematic of full-adder.

Inputs			Outputs	
A	B	Cin	Carry	Sum
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0

Truth table for Full adder

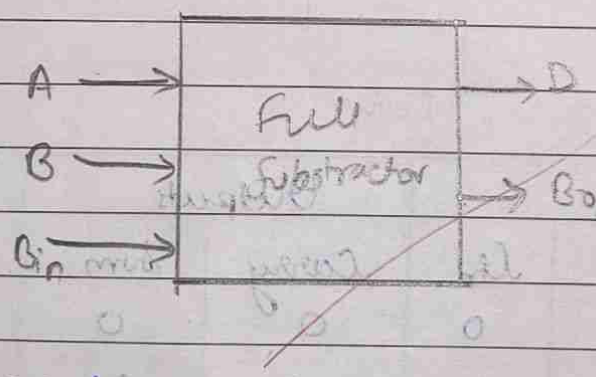
1	1	0	1	0
1	1	1	1	1

- **Logic Expressions.**

  - Sum:- $S = A \oplus B \oplus C_{in}$
  - Carry:- $C_{out} = AB + BC_{in} + AC_{in}$

- A full adder can be implemented using:
  1) Two Half Adders
  2) One OR gate.

- **Full subtractor:-**
  1) It is a combinational logic ckt designed to perform subtraction of three single bits.
  2) It contains three input (A, B, B_{in}) and produces two outputs (D, B_{out}).
  3) Where A & B are called Minuend and subtrahend bits.
  4) And, B_{in} → Borrow In & B_{out} → Borrow out.



- **Logic Expressions:**
  1) Difference:-

$$D = A \oplus B \oplus B_{in}$$

  2) Borrow:-

$$B_{out} = A'B + BB_{in} + A'B_{in}$$

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Inputs			Outputs	
A	B	Bin	D	Bout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

- A full subtractor can be implemented using:-
  1) Two half subtractors
  2) One OR gates

Q2) What do you mean by combinational circuit? Implement following Boolean function by using $F(A,B,C,D) = \Sigma m(0,2,4,6,7,9,13,15)$, $F(A,B,C,D) = \Pi M(0,3,4,6,8,9,13,14)$

Sol^n A combinational circuits is digital logic circuit in which output depends only on present input, it has no memory, No feedback is used, Output changes immediately when input changes.

1) $F(A,B,C,D) = \Sigma m(0,2,4,6,7,9,13,15)$

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